Non-isomorphic graphs have distinct ordinary and induced vertex-Ramsey classes

Maria Axenovich, Patrick Schmitteckert

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Allowed standard prerequisites (editorial).

- High-girth high-chromatic uniform hypergraphs. Original Construction 1 attributes existence to Erdos-Hajnal; native graph definitions and complete repetition lemma are included.. The target forbidden-block separation and recursive induced construction remain inside the selected proof.

1 Introduction

For a graph H and $k \in \mathbb{N}$, let the *k-colour vertex Ramsey class* for H, denoted $\mathcal{R}^k(H)$, be the set of all graphs G such that for any colouring of the vertices of G in k colours there is a subgraph isomorphic to H, that we call a *copy of* H, with all vertices of the same colour. Similarly, we define $\mathcal{R}^k_{ind}(H)$ if the copy of H in the above definition is an induced subgraph of G. The fact that $\mathcal{R}^k_{ind}(H)$ is nonempty for every graph H was first shown by Folkman [4], see also [1, 2, 6, 7] for results on vertex Ramsey theory. In this note we prove that any two non-isomorphic graphs have different k-colour Ramsey classes. Here, we assume that all graphs considered have at least one vertex.

Theorem 1.1. *For any two non-isomorphic graphs H_1 and H_2 and any positive integer k, $\mathcal{R}^k(H_1) \neq \mathcal{R}^k(H_2)$ and $\mathcal{R}^k_{ind}(H_1) \neq \mathcal{R}^k_{ind}(H_2)$.*

The study of Ramsey classes is a fundamental area in discrete mathematics addressing quantitative and structural properties of graphs in these classes. Most existing research on Ramsey classes has concerned the setting of edge colourings. A main motivation of this note is a question by Fox, Grinshpun, Liebenau, Person, and Szabó [5], whether any two connected non-isomorphic graphs have distinct edge Ramsey classes. While this question remains open in general, we answer the analogous question in the positive for vertex colourings and all, not only connected, graphs. In Section 2 we give some notations and needed Ramsey-type properties. Theorem 1.1 is proved in Section 3.

2 Notation and Ramsey properties

Let $H = (V, E)$ be a graph. We denote its order $|V(H)|$ by $|H|$, its size $|E(H)|$ by $||H||$, its complement by $\overline{H}$, and its independence number by $\alpha(H)$. If $||H|| \geq 1$, let $\beta(H)$ be the largest order of a block of H, that is, a maximal 2-connected subgraph or a bridge; if $||H|| = 0$, let $\beta(H) = 1$. For two graphs G and H, let $G \vee H$ denote their join, that is a graph obtained as a union of vertex-disjoint copies of G and H together with all possible edges between these two copies. We denote the set of all graphs with no induced copy of H by $\mathrm{Forb}(H)$.

Construction 1. Let H be a graph and q be an integer at least 3. Consider an $|H|$-uniform hypergraph $\mathcal{H}$ of girth at least q and chromatic number at least $k+1$. The existence of such a graph was first shown by Erdős and Hajnal [3]. Let $G = G(H, q, k)$ be a graph on vertex set $V(\mathcal{H})$ that is a union of $|E(\mathcal{H})|$ copies of H induced by respective hyperedges of $\mathcal{H}$. We have that $G \in \mathcal{R}^k_{ind}(H)$ since any vertex-colouring of G in k colours and thus of $\mathcal{H}$ results in a monochromatic hyperedge of $\mathcal{H}$, i.e., a monochromatic induced copy of H. Note that this copy is indeed induced since the girth condition ensures that any two hyperedges of $\mathcal{H}$ intersect in at most one vertex.

The definition of $\mathcal{R}^k_{ind}$ implies that $\mathcal{R}^k_{ind}(H_1) \neq \mathcal{R}^k_{ind}(H_2)$ if and only if $\mathcal{R}^k_{ind}(\overline{H}_1) \neq \mathcal{R}^k_{ind}(\overline{H}_2)$. Using Construction 1 with sufficiently large girth and complementation we see that for any graph H and any positive integer k, there is $G \in \mathcal{R}^k_{ind}(H)$ such that $\alpha(H) = \alpha(G)$.

For a graph A, we denote a pairwise vertex-disjoint union of q copies of A by qA. For $k \geq 1$, we say that a graph H has the *Ramsey repetition property* if for any integer $q \geq 1$ and every graph A, $qA \in \mathcal{R}^k_{ind}(H)$ if and only if $A \in \mathcal{R}^k_{ind}(H)$.

Lemma 2.1. *For any $k \geq 1$, a graph has the Ramsey repetition property if and only if it is connected.*

Proof. Fix $k \geq 1$.

Assume that H is connected. Let $q \geq 1$. If $A \in \mathcal{R}^k_{ind}(H)$, then clearly $qA \in \mathcal{R}^k_{ind}(H)$. If $A \notin \mathcal{R}^k_{ind}(H)$, repeat a colouring of A without a monochromatic H on its disjoint copies. This implies that $qA \notin \mathcal{R}^k_{ind}(H)$. Therefore H has a Ramsey repetition property.

Let H be disconnected and have components $H_1, \ldots, H_t$, where $t \geq 2$. Let $L = H_1 \vee \cdots \vee H_t$. Choose $A \in \mathcal{R}^k_{ind}(L)$ with $\alpha(A) = \alpha(L)$. Since $\alpha(A) = \alpha(L) = \max_i \alpha(H_i) < \sum_i \alpha(H_i) = \alpha(H)$, A contains no induced H, thus $A \notin \mathcal{R}^k_{ind}(H)$. Consider an arbitrary k-vertex colouring of qA, for $q = k(t-1)+1$. Each copy of A contains a monochromatic L, and t of these have the same colour. Taking H_i from the ith of these t distinct copies gives a monochromatic induced H. Therefore $qA \in \mathcal{R}^k_{ind}(H)$. Thus H has no Ramsey repetition property. □

3 Proof of Theorem 1.1

Let H_1 and H_2 be two non-isomorphic graphs. If $k = 1$, we have that $H_i \in \mathcal{R}^1_{ind}(H_i)$, $i \in \{1, 2\}$, thus $\mathcal{R}^1_{ind}(H_1) \neq \mathcal{R}^1_{ind}(H_2)$ and $\mathcal{R}^1(H_1) \neq \mathcal{R}^1(H_2)$. So, we assume that $k \geq 2$.

We shall first show that $\mathcal{R}^k(H_1) \neq \mathcal{R}^k(H_2)$. If $k = 1$, we have that $H_i \in \mathcal{R}^k(H_i)$, thus $\mathcal{R}^k(H_1) \neq \mathcal{R}^k(H_2)$. Let $k \geq 2$. Assume first that $|H_1| \neq |H_2|$. If $n = |H_1| < |H_2|$, let $G = K_{k(n-1)+1}$. By pigeonhole principle we see that $G \in \mathcal{R}^k(H_1)$. On the other hand, $G \notin \mathcal{R}^k(H_2)$ because one can assign k colours to the vertices of G such that each colour class has size at most n. Now, assume that $|H_1| = |H_2| = n$. Since H_1 and H_2 are not isomorphic, without loss of generality assume that H_1 is not a subgraph of H_2. Let $G = H_2 \vee K_{(k-1)(n-1)}$. To see that $G \in \mathcal{R}^k(H_2)$, consider an arbitrary vertex-colouring of G. At least n vertices are of the same colour, say red. Then the set of n red vertices induces a graph $G'' = G' \vee K_{n-\ell}$, for some $\ell \geq 0$, where G' is a subgraph of H_2 on ℓ vertices. Thus, G'' contains H_2 as a subgraph. On the other hand, by colouring the vertices of G inducing H_2 red and colouring the remaining vertices in other, $k-1$ colours, such that each of these colours is present on $(n-1)$ vertices, one has no monochromatic copy of H_1.

Next, we shall prove that $\mathcal{R}^k_{ind}(H_1) \neq \mathcal{R}^k_{ind}(H_2)$. Assume otherwise, i.e., that $\mathcal{R}^k_{ind}(H_1) = \mathcal{R}^k_{ind}(H_2)$. By Lemma 2.1 we see that both H_1 and H_2 are connected or both of them disconnected. By complementation, we can assume that both of them are connected. If one of H_1 or H_2, say H_1, is a single vertex graph, K_1, then $K_1 \in \mathcal{R}^k_{ind}(H_1) \setminus \mathcal{R}^k_{ind}(H_2)$, so we are done. We assume that H_1 and H_2 are non-isomorphic connected graphs each containing at least one edge.

Observe that if H is connected and has an edge, there is an independent set $I \subseteq V(H)$ such that $\beta(H - I) < \beta(H)$. Indeed, we can greedily select such an I by starting with some block as a root in the block-cutvertex tree and processing its blocks away from the root. Choose one vertex in the root block. For each subsequent block, if its attachment vertex has already been chosen, choose nothing; otherwise choose any other vertex in that block. A new vertex has no earlier neighbours except possibly the attachment vertex, so the chosen set I is independent and meets every block.

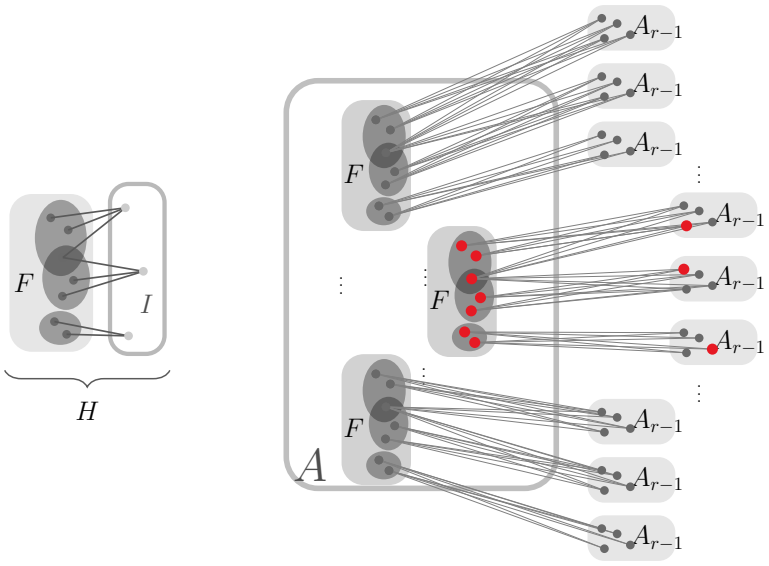


Figure 1: Construction of the graph A_r, $r \geq 2$ and a red copy of $H = H_2$.

Assume without loss of generality that $\beta(H_1) \geq \beta(H_2)$ and if $\beta(H_1) = \beta(H_2)$, then $|H_1| \geq |H_2|$. In particular $H_2 \in \text{Forb}(H_1)$. Choose an independent set I of H_2 such that for $F = H_2 - I$, $\beta(F) < \beta(H_2) \leq \beta(H_1)$.

First, we shall construct a graph $A \in \text{Forb}(H_1) \cap \mathcal{R}^k_{ind}(F)$. If F has no edges, let A be an edgeless graph on $k(|F| - 1) + 1$ vertices. If F has edges, then $\beta(F) \geq 2$, thus $\beta(H_1) \geq 3$. Let B be a block of H_1 on $\beta(H_1)$ vertices. Apply Construction 1 to build a graph $A = G(F, ||B|| + 2, k)$. Since the girth of the underlying hypergraph is at least $||B|| + 2$ and B is not contained in copies of F placed in hyperedges of that hypergraph, $A \in \text{Forb}(B)$, thus $A \in \text{Forb}(H_1)$. By the property of Construction 1, $A \in \mathcal{R}^k_{ind}(F)$. Thus $A \in \mathcal{R}^i_{ind}(F)$ for all $i \in \{2, \ldots, k\}$.

Next, we recursively define graphs A_r for $r \in \{1, \ldots, k\}$ such that $A_1 := H_2$ and for $r \geq 2$, $A_r \in \mathcal{R}^r_{ind}(H_2) \setminus \mathcal{R}^r_{ind}(H_1)$. For $r \geq 2$, we construct the graph A_r based on A_{r-1} as follows. Start with A. For every induced copy $\varphi(F)$ of F in A and every $u \in I$, add a copy $H_{\varphi,u,r-1}$ of A_{r-1}, such that these copies and A are all pairwise vertex-disjoint. Make each vertex of $H_{\varphi,u,r-1}$ adjacent to exactly those vertices of $\varphi(F)$ that correspond to the neighbours of u in H_2, see Figure 1.

We have that $A_r \notin \mathcal{R}^r_{ind}(H_1)$. Indeed, colour the vertices of A red and the remaining vertices in other $r-1$ colours such that each copy of A_{r-1} has no monochromatic induced copy of H_1. This colouring has no induced monochromatic copy of H_1.

On the other hand, $A_r \in \mathcal{R}^r_{ind}(H_2)$. To see this, consider an arbitrary r-colouring of $V(A_r)$. If some added copy $H_{\varphi,u,r-1}$ of A_{r-1} uses at most $r-1$ colours, it has a monochromatic induced copy of H_2. Otherwise, each such copy has all r colours on its vertices. Choose a monochromatic, say red, induced copy $\varphi(F)$ of F in A, and for each $u \in I$ choose a red vertex of $H_{\varphi,u,r-1}$. Together these vertices and $\varphi(F)$ induce a red copy of H_2.

In particular, $A_k \in \mathcal{R}^k_{ind}(H_2) \setminus \mathcal{R}^k_{ind}(H_1)$, implying that $\mathcal{R}^k_{ind}(H_1) \neq \mathcal{R}^k_{ind}(H_2)$ and contradicting our assumption. □

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